ALGEBRA 2 · UNIT 11: PROBABILITY & STATISTICS

Carnival Game Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design games for a school carnival. They count outcomes with permutations and combinations, find probabilities of independent and dependent events, use two-way tables for conditional probability, compute expected value, use the normal distribution and binomial probabilities, and design a booth that raises money.

Expected value connects probability to decision making. Students see why carnival games are designed slightly in the house's favor and design a game with a target expected value.

Standards Addressed

- S-CP.2–5 Independence and conditional probability
- S-CP.9 Permutations and combinations
- S-MD.2–4 Expected value
- S-ID.4 Normal distribution

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: play Lucky 7 for 10 rounds. Part 1 probability and counting.
Day 2	Part 2 expected value and distributions.
Day 3	Part 3 design and test.

Materials

- Student handout
- Dice, cards, or spinners
- Calculators
- Optional: a carnival booth trial run

How the Three Versions Differ

Level	What changes
CP2	Simple and compound probability, counting with formulas, and expected value with a table.
CP1	Conditional probability from a two-way table, dependent events, the normal distribution, and a target expected value.
Honors	Binomial probabilities, the variance of profit, and a design with simulation and a fairness argument.

Watch For (Common Misconceptions)

- Treating draws without replacement as independent.
- Forgetting to subtract the cost to play.
- Mixing up permutations and combinations.

Extension Ideas

- Run the booth at a real event.
- Simulate 1,000 plays with a spreadsheet.
- Research casino house edges.

Answer Key – CP2

Part 1: Lucky 7 & Ducks

#	Task	Answer
1	List the ways to roll a sum of 7. What is the probability of winning?	(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1): $6 \div 36 = \frac{1}{6}$
2	Duck pond: 4 of 10 ducks are winners. A player picks 2 ducks without putting the first back. Find the probability both are winners.	$(4 \div 10)(3 \div 9) = 12 \div 90 = 2 \div 15 \approx 0.133$
3	How many ways can 3 prizes be chosen from 10 different prizes? (combinations)	$C(10, 3) = 120$
4	How many ways can 3 of 5 prizes be arranged in a row on a shelf? (permutations)	$P(5, 3) = 5 \times 4 \times 3 = 60$

Part 2: Expected Value

#	Task	Answer
5	Complete the table for the player's net gain and find the expected value.	$\frac{1}{6}$ and $\frac{5}{6}$; $EV = 4(\frac{1}{6}) - 1(\frac{5}{6}) = -\frac{1}{6} \approx -\$0.17$
6	If 300 people play, about how much money does the booth make?	$300 \times \$0.17 \approx \$50$
7	Ring toss: P(win) = 0.15, costs $2, pays $10. Find the player's expected value.	$0.15(8) + 0.85(-2) = 1.20 - 1.70 = -\$0.50$

Part 3: Design a Booth

#	Task	Answer
8	Describe the game and find the probability of winning.	Answers vary.
9	Choose the cost and prize. Find the expected value for the player.	Answers vary.
10	Predict the earnings for 200 plays.	Answers vary.

Answer Key – CP1

Part 1: Who Wins?

#	Task	Answer
1	Use the table to find P(win), P(win | adult), and P(win | student). Are winning and age independent?	P(win) = 34 ÷ 200 = 0.17; P(win | adult) = 14 ÷ 80 = 0.175; P(win | student) = 20 ÷ 120 ≈ 0.167. Very close, so they are approximately independent.
2	Duck pond: 4 winners of 10. Find P(exactly one winner) when picking 2 ducks without replacement.	$2 \times (4 \div 10)(6 \div 9) = 48 \div 90 = 8 \div 15 \approx 0.533$
3	A spinner has 8 equal sections, 2 of them gold. Find P(gold on both of two spins). Are the spins independent?	$(¼)(¼) = 1 \div 16 = 0.0625$; yes, each spin doesn't affect the other.

Part 2: Expected Value & Normal

#	Task	Answer
4	Lucky 7 costs $1. What prize makes the player's expected value exactly −$0.25?	$(⅛)(x - 1) + (⅞)(-1) = -0.25 \rightarrow x \div 6 - 1 = -0.25 \rightarrow x = \$4.50$
5	The time to play a game is normally distributed with mean 4 min and standard deviation 1 min. What percent of games last between 3 and 5 min? More…	About 68% between 3 and 5 min; about 2.5% more than 6 min (2 standard deviations).
6	How many games can the booth run in a 3-hour shift, on average?	$180 \div 4 = 45$ games

Part 3: Design a Booth

#	Task	Answer
7	Write the rules and the probability model.	Answers vary.
8	Choose a cost and prize to hit the target exactly. Show the equation.	Answers vary.
9	Predict earnings for a 3-hour shift.	Answers vary.

Answer Key – Honors

Part 1: Repeated Plays

#	Task	Answer
1	Find the probability of winning at least once.	$1 - (\frac{5}{6})^5 \approx 0.598$
2	Find the probability of exactly 2 wins.	$C(5, 2)(\frac{1}{6})^2(\frac{5}{6})^3 \approx 0.161$
3	Find the expected number of wins in 5 plays.	$5 \times \frac{1}{6} \approx 0.83$ wins

Part 2: Profit & Risk

#	Task	Answer
4	Find the expected booth profit per play and for 300 plays.	$\mu = \frac{1}{6}(-4) + \frac{5}{6}(1) = \frac{1}{6} \approx$ \$0.17 per play; about \$50 for 300 plays
5	Find the standard deviation of the booth's profit per play. (Values +\$1 or −\$4.)	Variance $= \frac{1}{6}(-4 - \frac{1}{6})^2 + \frac{5}{6}(1 - \frac{1}{6})^2 \approx 2.894 + 0.579 = 3.472$; $\sigma \approx$ \$1.86
6	Use the normal approximation: about how likely is the booth to lose money over 300 plays? (Total profit has mean 300μ and standard deviation √300…	Mean \$50, SD $\approx \sqrt{300} \times 1.86 \approx$ \$32.27. $z = -50 \div 32.27 \approx -1.55$: about 6% chance of losing money.

Part 3: Design & Simulate

#	Task	Answer
7	Write the probability model and expected value.	Answers vary.
8	Simulate 50 plays (dice, cards, or a random-number generator). Compare with your model.	Answers vary.
9	Write the proposal with the risk analysis.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Designed game has a correct probability model, and its expected value meets the target. Goes beyond the minimum with a creative, realistic choice.	Designed game has a correct probability model, and its expected value meets the target.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes the game rules, probabilities, expected value per play, and predicted earnings. Polished, organized, and visually engaging.	Proposal includes the game rules, probabilities, expected value per play, and predicted earnings. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.